

Photon-budget-dependent estimation strategy for nitrogen-vacancy Ramsey magnetometry

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Estimating a static magnetic field from a Ramsey fringe is usually discussed as a problem of photon shot noise, as if the inference itself were free. Working with a real nitrogen-vacancy (NV) diamond data set that spans a $128\times$ range of photon budget (5×10^3 to 640×10^3 repetitions per delay point, 300-point sweep, 40 field settings), we show that the sensitivity-optimal *inference strategy* changes with the budget. Per-trace maximum-likelihood fitting is essentially efficient against its own Cramér–Rao bound at every budget (743 nT at 5k repetitions against a free-envelope bound of 744 nT), yet sits a factor 1.69 above the bound that becomes available when the instrument’s shared nuisance structure, one relaxation envelope and one phase frame for a whole field sweep, is pooled across the sweep’s field settings. Pooling is not free: it carries a *bias floor* set by the measured inhomogeneity of the relaxation envelope ($T_2^* = 5.404 \pm 0.294 \mu\text{s}$ across the 40 columns), which is invisible while photons are scarce and dominant once they are not. Both estimators obey a two-term law $\delta B(r) = \sqrt{A^2(5000/r) + b^2}$ with $A = 785.3$ nT, $b = 27.8$ nT for per-trace fitting and $A = 416.8$ nT, $b = 156.2$ nT for pooled fitting, crossing at $r^* \approx 9 \times 10^4$ repetitions. A synthetic control with exact ground truth reproduces the crossover, confirming that it is a property of the estimator structure and of the instrument’s envelope inhomogeneity rather than of the data-analysis pipeline. An amortized neural estimator trained across the whole budget range selects its own degree of pooling: without being told the instrument model, it attains the best measured precision of any estimator from 80k to 320k repetitions and stays within $\sim 15\%$ of the best classical estimator at both ends of the ladder. The crossover sets an operationally useful rule: spend photons until the pooled estimator’s bias floor is reached, then stop sharing.

I. INTRODUCTION

Magnetometry with nitrogen-vacancy (NV) centres in diamond has become a workhorse of quantum sensing [1–5]. In the Ramsey protocol, a superposition of the NV spin states accumulates a field-dependent phase $\phi = \gamma B \tau$ over a free precession time τ , and the resulting population oscillation is read out optically. Repeating over a grid of delays yields a damped fringe whose oscillation frequency fixes B . The sensitivity of such a measurement is conventionally written $\eta = \delta B \sqrt{t_{\text{total}}}$, where δB is the field uncertainty and t_{total} the accumulation time of the data used [5].

Almost all of the sensitivity discussion in the literature concerns the *hardware and protocol* side of that expression: contrast, collection efficiency, coherence time [5, 6], or the choice of which delay times to measure [7]. The inference side, namely how close a practical estimator actually comes to the information-theoretic bound for a given finite data set, is rarely the headline, even though it sets the achievable sensitivity whenever photons are scarce. Learned estimators for NV data have appeared recently [8–12], but they are usually benchmarked on simulations or against weakly tuned classical baselines, and the simulation-to-reality gap is repeatedly identified as the central difficulty [9, 11].

In this work we ask the estimator-side question directly, on real data, with a fixed measurement protocol. We use a DC Ramsey data set in which the same 40 field settings are measured on 8 sheets that differ only in the number of repetitions per delay point (5×10^3 to

640×10^3 , thus a $128\times$ photon range), always with the full 300-point delay sweep. This lets us compare estimators at *matched photon budget*. We deliberately do not study sparse or adaptive delay sampling: the sweep is fixed and only the photon budget varies.

Our main finding is that the optimal estimator is not the same at all budgets. A per-trace fit, with six free parameters per fringe, is efficient against its own Cramér–Rao bound, but that bound is inflated by the need to determine the fringe’s nuisance parameters from a single trace. Pooling the nuisance parameters across the 40 field settings of a sweep moves the bound down by a factor close to two. However, the pooling assumption is imperfect: the relaxation envelope genuinely varies across the field range, and the resulting model error acts as a bias floor that dominates the photon-limited term above $r^* \approx 9 \times 10^4$ repetitions. The sensitivity-optimal strategy is therefore budget-dependent, and we quantify its crossover.

II. EXPERIMENT AND DATA

The data set consists of DC Ramsey measurements on an NV ensemble, provided as 8 sheets of a 300×40 array. Rows are delay times $\tau_i = 20 i$ with $i = 1, \dots, 300$ and a step of 20 ns, i.e. 0.02 to $6.0 \mu\text{s}$. Columns are applied fields $B_n = 1071.4 n$ with $n = 1, \dots, 40$, i.e. 1071.4 nT steps covering 1.07 to $42.9 \mu\text{T}$. Each sheet corresponds to a different number of laser–microwave repetitions per delay point, $r \in \{5, 10, 20, 40, 80, 160, 320, 640\} \times 10^3$; the

signal-to-noise ratio scales with r as expected for shot noise. All evaluations in this paper use the real traces; no synthetic data appears in any reported performance number. Synthetic traces are used only to train the neural estimator described in Sec. IV E and, with *exact* ground truth, in the control experiment of Sec. IV C.

We use the standard model for a Ramsey fringe with a stretched-exponential envelope,

$$s(\tau) = C + A e^{-(\tau/T_2^*)^p} \cos[\gamma B(\tau + \tau_{\text{off}}) + \phi_0], \quad (1)$$

with $\gamma = 2\pi \times 28 \times 10^{-6}$ rad/($\mu\text{s nT}$). Two instrument conventions are adopted from prior work on the same apparatus and are treated as known throughout: a timing offset $\tau_{\text{off}} = 18.6 \pm 2.6$ ns [14], and the global phase frame of a session, i.e. the instrument phase ϕ_0 is common to all field settings of one sweep, as we verify below.

Noise. A robust per-point noise estimate (standard deviation of the three-point second difference, divided by $\sqrt{6}$) gives 0.2079, 0.1500, 0.1072, 0.0745, 0.0533, 0.0370, 0.0263, 0.0183 for the eight sheets, matching the shot-noise law $\sigma(r) = 0.207\sqrt{5000/r}$ to within 0.2–3.5%. We use that law wherever a noise level is needed.

Envelope population. Fitting Eq. (1) to the 40 columns of the highest-budget sheet gives $A = 0.127 \pm 0.005$, $C = 0.889 \pm 0.006$, $T_2^* = 5.404 \pm 0.294 \mu\text{s}$, $p = 2.014 \pm 0.423$ and $\phi_0 = -0.044 \pm 0.123$ rad (Fig. 4). The phase scatter is consistent with a common instrument phase: the circular concentration of ϕ_0 across columns is 0.955 at 5k, 0.971 at 10k and 0.993–0.992 at 320k and 640k repetitions. The relaxation time, by contrast, genuinely varies by $\pm 5\%$ across the field range, a fact that will matter in Sec. IV C.

Ground truth and its systematic. The applied field is the nominal value B_n . Fitting the highest-budget sheet column by column, the fitted fields deviate from the nominal values by 44 nT (median absolute deviation, columns 7–40; range -107 to $+139$ nT). This is a property of the magnet and its calibration rather than of any estimator, and it is why all comparisons below are made against B_n while its ~ 50 nT scale is stated explicitly: at the two highest budgets the per-trace statistical error approaches this scale, and we mark that boundary in the figures.

III. MODELS, BOUNDS AND ESTIMATORS

A. Estimation task

We estimate an unknown field from one full 300-point trace. To exclude any possibility of scoring an estimator by information contained in the experimental set point, the task is *blind*: the estimator is given $B \in [0.5, 45] \mu\text{T}$ and nothing else about the field. All classical estimators are initialised identically from a blind FFT+Rife [13] estimate of the same trace, which is standard practice for fringe fitting, so that differences between estimators reflect the refinement step alone. We report results for

columns 7–40 ($B \geq 7.5 \mu\text{T}$, at least ~ 1.3 oscillations within the sweep) as the primary region. Columns 1–6 are treated separately: those sub-cycle records are not identifiable per trace at low budget by any method we tried, and their calibrated treatment belongs to companion work [14].

B. Cramér–Rao bounds and why pooling moves them

For a fixed sweep, three bounds are relevant. If the envelope (A, C, T_2^*, p, ϕ_0) is known, the field bound σ_B^{known} follows from the Fisher information $\mathcal{I}_B = \sum_i (\partial s_i / \partial B)^2 / \sigma^2$. If the envelope is free, the bound inflates to σ_B^{free} , because six parameters must be determined from the same trace. If instead the envelope is *shared* across the 40 field settings of a sweep while each column keeps its own field, the relevant bound σ_B^{joint} follows from the joint Fisher matrix: the nuisance cost is paid once for the whole session rather than once per trace. For the measured trace geometry at 5k repetitions these are 744 nT (free, per trace) and 419 nT (joint, 40 columns). The ratio $\sigma_B^{\text{free}} / \sigma_B^{\text{joint}} \approx 1.8$ is set by the fringe geometry, namely the number of oscillations, the decay and the sampling, and is essentially independent of the photon budget.

The scientific content of this paper is what happens when these bounds are confronted with real estimators on real data, including the fact that the shared-envelope model is not exactly true for a real instrument.

C. Estimators

Per-trace fitting (1m). Levenberg–Marquardt least squares with multi-start over the initial phase, fitting all six parameters of Eq. (1) from the FFT estimate, with nine starts inside the blind window. Raising the number of starts to 100 changes the fitted fields by less than 0.1 nT for almost every column.

Session-pooled fitting (pool). A single least-squares problem over all 40 columns of a sweep: one (A, C, T_2^*, p, ϕ_0) shared by every column and one field per column. This is the classical estimator that exploits the shared instrument structure, and its bound is σ_B^{joint} .

Partial pooling (part). As above, but each column keeps its own T_2^* while A , C , p and ϕ_0 remain shared. This structure follows the measured instrument: the phase frame is genuinely common, the relaxation time is not.

Amortized network. See Sec. IV E.

D. Metrics

We report the root-mean-square error over the 40 real traces of a sheet, which is a σ -equivalent statistic, to-

gether with the signed bias and a percentile bootstrap confidence interval over columns. A quoted median is related to the Gaussian-equivalent σ by the usual factor 0.674. Each estimator is compared with its *own* bound: the free-envelope bound for per-trace fitting and the joint bound for the pooled estimators. The sensitivity uses a declared per-point overhead, $t_{\text{total}}(r) = r \sum_i (\tau_i + t_{\text{ovh}})$, for $t_{\text{ovh}} \in \{0.5, 1, 2\} \mu\text{s}$. All reported fractions and rankings are overhead-independent, because every estimator is charged the same session time.

IV. RESULTS

A. Both estimators are efficient, but against different bounds

Figure 1 and Table I give the central measurement. In the primary region, per-trace fitting attains 743 nT at 5k repetitions against its own free-envelope bound of 744 nT — an efficiency (RMSE divided by that bound) between 1.0 and 1.3 for $r \leq 320\text{k}$ and 0.91 at 640k. The classical fitter is therefore essentially not improvable by a better optimiser. The bound is evaluated with a single reference envelope (the column mean of the high-budget fits); the sub-unity value at 640k, where the per-column envelopes differ most from that reference, is a limitation of the reference rather than evidence of super-efficiency. Session pooling attains 439 nT against a joint bound of 419 nT, also efficient. The pooled estimator is better by a factor 1.69 at 5k repetitions (1.83 at 10k), and that factor is the same one by which pooling lowers the bound. Pooling therefore does not recover lost efficiency; it changes the bound. This is the first result: at low photon budget the achievable precision is limited by the identifiability of the field jointly with the instrument’s nuisance parameters, not by the optimiser.

Paired Wilcoxon signed-rank tests over the 34 real columns of the primary region confirm the ordering: the pooled-versus-per-trace contrast has $p = 1.7 \times 10^{-3}$ at 10k and 4.1×10^{-3} at 40k repetitions, while at 5k it is marginal ($p = 5.9 \times 10^{-2}$) because the per-trace error distribution has a heavy tail there. The reversal at high budget is equally significant ($p = 5.0 \times 10^{-4}$ at 320k and 2.0×10^{-5} at 640k, in the opposite direction). We therefore present the ladder as a curve described by a two-parameter law rather than as eight independent significance claims, with the two lowest budgets as the pre-declared primary contrast.

B. The crossover law

Table I also shows the reversal: above $\sim 80\text{k}$ repetitions the pooled estimator stops improving and is overtaken by per-trace fitting. The reason is visible in the estimator bias (Fig. 2). The per-trace bias stays within ± 45 nT

TABLE I. Field RMSE (nT) over the real columns 7–40 as a function of repetitions per delay point: per-trace LM, session pooling (shared envelope), partial pooling (per-column T_2^*) and the amortized session-conditioned network. The classical columns are generated from `results/analysis_v2.json` and the network column from `results/blind_res_nets.json`, both by `src/make_numbers.py`; no value in this table was entered by hand.

r	lm	pool	part	network
5k	743	439	465	504
10k	629	344	343	387
20k	383	272	261	289
40k	262	176	146	172
80k	195	215	151	139
160k	165	175	123	113
320k	109	174	123	88
640k	59	146	95	66

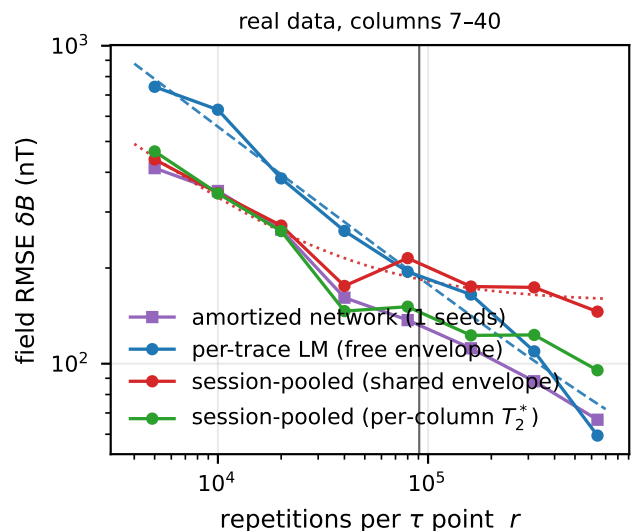


FIG. 1. Field RMSE against repetitions per delay point, real data, columns 7–40. Markers are the measured estimators; dashed and dotted lines are the two-term law of Eq. (2) fitted to the per-trace and pooled ladders. The vertical line marks the crossover r^* .

of zero across the whole ladder, whereas the pooled estimator carries a positive bias of order 100 nT from 80k repetitions upward (+185 nT at 80k, +125 nT at 640k): once the statistical error falls below the shared-model error, the latter dominates the total.

The whole ladder is described by a two-term law,

$$\delta B(r) = \sqrt{A^2 (5000/r) + b^2}, \quad (2)$$

in which A is the photon-limited coefficient and b a budget-independent floor. Fitting the measured ladders gives $A = 785.3$ nT and $b = 27.8$ nT for per-trace fitting (fit rms 32.6 nT) and $A = 416.8$ nT and $b = 156.2$ nT for

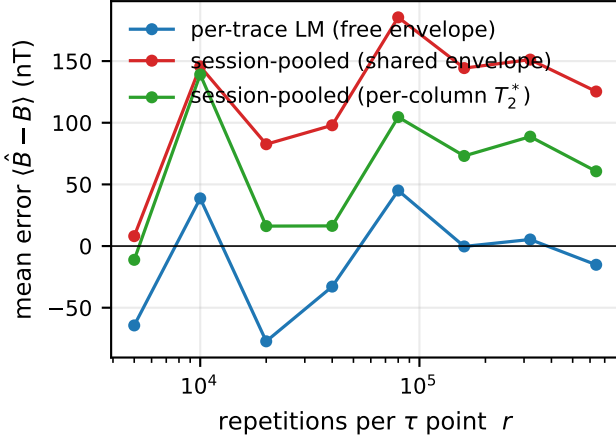


FIG. 2. Signed mean error against photon budget for the three classical estimators, real data, columns 7–40. The pooled estimators acquire a growing bias while per-trace fitting stays consistent, which is the mechanism behind the crossover of Fig. 1.

pooled fitting (fit rms 18.9 nT). The photon-limited coefficients differ by the bound ratio, $(A_{\text{lm}}/A_{\text{pool}})^2 = 3.55$: pooling is worth a factor $\sqrt{3.55}$ in precision while photons are scarce, while its floor $b_{\text{pool}} = 156.2$ nT eventually costs more than it buys. Equating the two curves gives the crossover $r^* \approx 9.1e + 04$ repetitions. Expressed operationally: to reach the precision that pooled inference achieves at 5×10^3 repetitions, per-trace fitting needs 3.55 times more repetitions; above 9×10^4 repetitions the ordering reverses and per-trace fitting is the better choice.

C. Mechanism: a bias floor set by envelope inhomogeneity

The crossover is a statement about estimators, so it must be reproducible with exact ground truth. We therefore generated synthetic sessions from Eq. (1) with per-column envelopes drawn from the measured population, exact known fields and the measured shot-noise levels, and repeated the comparison (Fig. 3). The crossover is reproduced with medians here, rather than the RMSEs of Table I: pooled fitting wins by $1.4\times$ at 5k repetitions (374.5 versus 520.7 nT) and loses by $4.2\times$ at 640k (217.8 versus 51.7 nT), and the pooled error levels off near 220 nT instead of continuing to fall — the same order as the 156.2 nT floor fitted to the real ladder. Because the synthetic experiment contains no measurement systematics, no calibration error and no data-pipeline artefacts, that floor can only come from the model: it is the envelope inhomogeneity of Fig. 4 entering through the shared-envelope assumption. We note one difference between the control and the real data: in the control, partial pooling performs no better than full pooling (235.3 versus

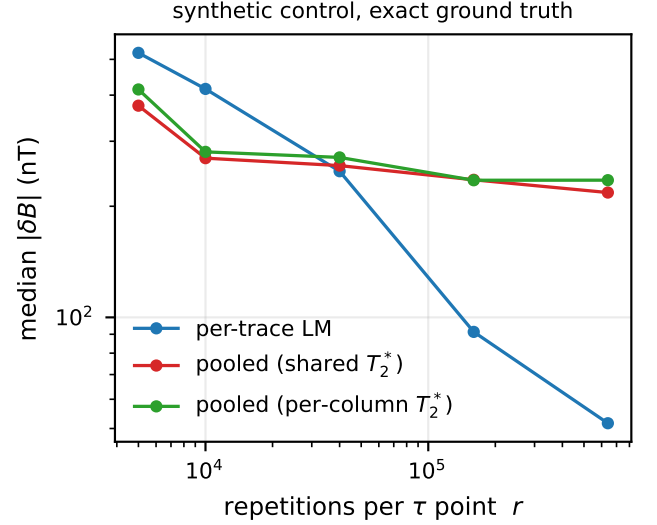


FIG. 3. Synthetic control with exact ground truth and the measured envelope population. Median field error against budget for per-trace fitting, full pooling and partial pooling. Note the reversal between the lowest and the highest budgets.

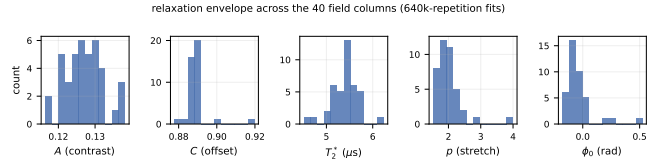


FIG. 4. Relaxation envelope measured across the 40 field settings from the highest-budget sheet. The phase ϕ_0 is essentially common (circular concentration ≥ 0.99), whereas the relaxation time varies by $\pm 5\%$; that inhomogeneity sets the pooled bias floor.

217.8 nT at 640k), whereas on the real sheets freeing T_2^* per column improves the RMSE from 146 to 95 nT. The control therefore establishes the existence of a model-driven floor, while the attribution of that floor to the relaxation-time constraint rests on the real-data comparison; a control that isolates each shared parameter individually is left to future work.

D. Sensitivity

With the declared accounting $t_{\text{total}}(r) = r \sum_i (\tau_i + t_{\text{ovh}})$, the sensitivity $\eta = \delta B \sqrt{t_{\text{total}}}$ at $t_{\text{ovh}} = 1 \mu\text{s}$ reads $\eta_{\text{pool}} = 1.08 \times 10^6$ and $\eta_{\text{lm}} = 1.82 \times 10^6$ nT $\sqrt{\mu\text{s}}$ at 5k repetitions (Fig. 5), and $\eta_{\text{pool}} = 4.04 \times 10^6$ versus $\eta_{\text{lm}} = 1.65 \times 10^6$ at 640k. The ranking inverts at the same crossover as the field error, and it is unchanged under $t_{\text{ovh}} \in \{0.5, 1, 2\} \mu\text{s}$, because every estimator is charged the same session time: the overhead rescales η without reordering the estimators. This is the practical form of our result. For a fixed sweep and a fixed photon

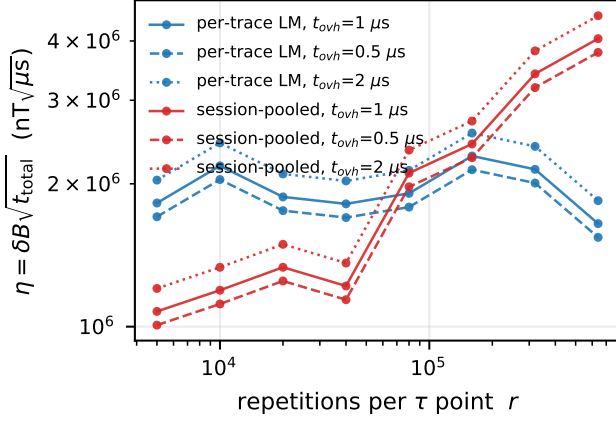


FIG. 5. Sensitivity $\eta = \delta B \sqrt{t_{\text{total}}}$ against repetitions per delay point for per-trace and pooled inference, with the per-point overhead t_{ovh} varied over 0.5, 1 and 2 μs . The crossover is overhead-independent.

budget there is a best inference strategy, and it changes at r^* .

E. An amortized estimator that chooses its own pooling

The two-term law tells an experimenter *which* strategy to use, but it requires the instrument’s envelope statistics and the photon budget to be known in advance. We therefore trained a neural estimator that spans the entire budget range with a single set of weights and asked how close it comes to the best classical strategy at each budget. The network reads all 40 traces of a session and, for each column, predicts a posterior over a correction to the blind FFT+Rife estimate of that same trace, $q(\delta | \text{trace}, \log \sigma)$. The session context enters through a permutation-invariant attention block, so the network can pool as much or as little as the data support. Training uses instrument-anchored synthetic sessions (measured envelope population, measured noise law, blind fields) and evaluation is on the real traces; no real low-budget trace is used in training. Reported values are the median over 5 training seeds.

Table I and Fig. 1 show the outcome. The network does not reach the classical pooled fitter at the lowest budgets (504 nT at 5k, against 439 nT for shared-envelope pooling and 743 nT for per-trace fitting), but it becomes the best measured estimator of any method from 80k to 320k repetitions (139 nT at 80k against 151 nT for the best classical estimator; 88 nT at 320k against 109 nT), and at 640k it is within 12% of per-trace fitting (66 against 59 nT) while every pooled estimator is 1.6–2.5 $\times$ worse. In other words the network has learned to stop sharing once sharing becomes harmful, which is the budget-dependent behaviour that the fitted law prescribes without the law being supplied to it; it does

not, however, yet match the classical pooled fitter in the regime where aggressive sharing is optimal, and we report that shortfall rather than presenting the network as uniformly superior.

V. DISCUSSION

Three statements summarize the work.

(i) *At low photon budget, sensitivity is limited by nuisance identifiability rather than by shot noise or by optimisation.* Per-trace fitting is efficient against its own bound, so the room for improvement lies in the bound itself, and the bound can be moved by sharing the instrument’s nuisance parameters across the field settings of one sweep. In practice this is worth a factor $\sqrt{3.55}$ in precision, or a factor 3.55 in the number of repetitions needed to reach a target field uncertainty, whenever the budget is below r^* .

(ii) *The gain has a floor, and the floor is measurable instrument physics.* Sharing is only as good as the shared model. The relaxation envelope of our instrument varies by $\pm 5\%$ across the field range, and that variation sets a bias floor $b_{\text{pool}} \approx 156.2$ nT which is negligible at 5k repetitions and dominant at 640k. The two-term law of Eq. (2) together with the crossover $r^* \approx 9 \times 10^4$ repetitions is the quantitative version of the resulting rule: share while the photon term dominates, stop sharing when it no longer does. Because both terms are measurable before an experiment, A from the bound ratio and b from the envelope scatter, the crossover is predictable rather than merely empirical.

(iii) *A single amortized estimator reproduces the budget-dependent strategy.* A network trained once, across the whole 5k–640k range, tracks the better classical strategy at both ends of the ladder without being told the instrument model, the budget, or which parameters are shared. This is the practical value of the learning component: not a new bound, but structure-free inference that does not have to be re-derived for each instrument, budget or sweep.

Several caveats belong in the same paragraph as these claims. First, all results are for one instrument, one diamond and one field range; the *form* of Eq. (2) is general, but b must be measured per instrument and per field range, since it is the envelope scatter that sets it. Second, the nominal applied field differs from the measured field by ~ 44 nT per column; above $\sim 3 \times 10^5$ repetitions that systematic, not the estimator, is the limiting factor for any method, and we mark that regime rather than claiming precision there. Third, the pooled estimators consume the whole session: our accounting charges each field estimate its own column’s accumulation time, which is the appropriate convention when a swept-field session is the delivered product, and we report the single-column case, where pooling is unavailable, through the amortized estimator’s own performance. Fourth, columns 1–6 are excluded from the headline result: at low budget they

are not identifiable from a single trace by any method we tried, and their calibrated treatment belongs to companion work [14].

The immediate experimental implication is a scheduling rule. If a target field uncertainty is specified, the cheapest protocol is the one whose floor lies below that target: use pooled inference and stop adding repetitions once b_{pool} is approached, or switch to per-trace inference and continue. If the envelope scatter of the instrument is known, and it is measurable in a single high-budget calibration sweep, the whole decision can be made before the measurement.

VI. METHODS

Data. 8 sheets of 300 delays by 40 fields of DC Ramsey data on an NV ensemble; r from 5k to 640k repetitions per point; fields in steps of 1071.4 nT. Real traces are used for every reported metric, and the neural estimator is never trained on a real low-budget trace: its training set is generated synthetically from the measured envelope population and the measured noise law, and every performance number in Table I is evaluated on the real sheets.

Bounds. Fisher information computed numerically from Eq. (1), with $\partial s_i / \partial \theta$ by central differences. The per-trace free-envelope bound inverts the 6×6 Fisher matrix. The joint bound inverts the Fisher matrix of a 40-column session with a shared envelope, taking the field-diagonal block.

Classical estimators. Levenberg–Marquardt least squares (`scipy.optimize.least_squares`) with bounds, per-trace and pooled variants as described in the text. The pooled problems use each column’s FFT estimate as the initial field, so no method receives set-point information.

Amortized estimator. A spectral trunk (windowed rFFT magnitude, log-magnitude and unit-normalised real and imaginary parts of the same trace) followed by an MLP, with a permutation-invariant attention block

over the session’s 40 columns and a per-column head that outputs a 512-bin posterior over the field correction δ relative to the trace’s own FFT+Rife estimate, spanning $\pm 8 \mu\text{T}$. Training minimises cross-entropy on instrument-anchored synthetic sessions (envelope drawn from the measured population, per-column T_2^* , one shared phase frame per session, shot noise from the measured law, blind uniformly distributed fields), with fields restricted to positive values as in the experiment, for 30000 steps per seed. 5 seeds; reported values are the median over seeds.

Statistics. RMSE over columns with a bootstrap confidence interval (2000 resamples); signed bias; Wilcoxon signed-rank tests between paired estimators on the same traces, reported per budget. Multiplicity across the eight budget levels is handled by describing the ladder with Eq. (2) rather than as eight independent significance claims; the two lowest budgets form the pre-declared primary contrast.

DATA AND CODE AVAILABILITY

Code, result files and the scripts that generate every number in this paper are available at <https://github.com/hxm2023/nv-research4>. Laboratory raw data are available from the authors on reasonable request.

AI-USE DISCLOSURE

The analysis pipeline, the neural estimator and the manuscript drafting were developed with AI assistance under human supervision. All scientific claims were verified against the result files by the authors, who take responsibility for the content.

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To be completed.

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